



APPLICATION OF RSA CRYPTOSYSTEM

BY - GROUP I

CSE – 3030 DISCRETE MATHEMATICS

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MEMBERS AND RESPONSIBILITIES

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2	2022-MIIT-CSE-007	Shin Thant Aung	Research & Calculation
3	2022-MIIT-CSE-014	Kyaw Zay Ya Hein	Report
4	2022-MIIT-CSE-021	Shon Lei Zaw	Presentation
5	2022-MIIT-CSE-028	Pyae Mon Thant	Report
6	2022-MIIT-CSE-034	Bhone Myint Maung	Research & Calculation
7	2022-MIIT-CSE-044	Aung Chan Myae	PowerPoint
8	2022-MIIT-CSE-050	Thadar Thant	PowerPoint
9	2022-MIIT-CSE-056	Yin Myat Noe Oo	Research & Calculation
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INTRODUCTION

Cryptography is the method of protecting information so that only the right people can read it. In simple terms, it changes normal readable data, called plaintext, into a scrambled form called ciphertext. Later, it can be turned back into the original message when needed. For a long time, most systems used the same secret key for both locking and unlocking the message. The biggest problem with this method was how to safely share that secret key between two people.

In 1976, a breakthrough happened with the invention of the RSA cryptosystem by Ron Rivest, Adi Shamir, and Leonard Adleman. RSA was the first practical public-key system. Instead of one secret key, it uses two keys: a public key (which everyone can know) and a private key (which is kept secret). This solved the key-sharing problem and made secure communication over the internet possible.

Today, RSA is everywhere. It is used for secure websites, online banking, digital signatures, email protection, and many other applications that keep our digital world safe. Because of its importance, RSA is often called the foundation of modern internet security.

Public Key vs. Private Key Cryptosystem

Feature	Public-Key Cryptosystem	Private-Key Cryptosystem
Encryption Key	Publicly known and distributed	Kept private and must be securely shared between communicating parties.
Decryption Key	Kept secret (the private key). Cannot be derived from the public key in any feasible time.	Same as the encryption key (shared secret key).
Primary Security	Security is based on hard mathematical problems such as factoring large primes and solving discrete logarithms	Security is based on the complexity of the cryptographic algorithm and secrecy of the key.
Key Distribution	Solves the key distribution problem; no need to securely share a secret key beforehand.	The major challenge: Requires a secure channel to distribute the secret key safely.
Speed & Efficiency	Slow and computationally intensive; not suitable for encrypting large volumes of data.	Very fast and optimized for bulk data encryption such as files, disks, and streaming
Cryptographic Method	Asymmetric Encryption	Symmetric Encryption
Number of Keys	Two keys (one public, one private).	One shared key.
Primary Use Cases	Key exchange, digital signatures (authentication and non-repudiation), encrypting small data.	Bulk data encryption, message authentication codes (MACs), secure file and database storage.
Examples	RSA, Elliptic Curve Cryptography (ECC), Diffie-Hellman, ElGamal, DSA.	AES, DES, 3DES, Blowfish, Shift cipher, Affine cipher.

Mathematical Background

For RSA: Public encryption key: (n, e)

Private decryption key: (d, n)

RSA encryption is done by the function:

$$C = M^e \bmod n$$

RSA decryption is done by the function:

$$M = C^d \bmod n$$

n is obtained by choosing two large primes p and q:

$$n = p \cdot q$$

Next, calculate $\phi(n) = (p-1)(q-1)$

Choose an exponent e that is relatively prime to $\phi(n)$:

$$\gcd(e, \phi(n)) = 1$$

This implies that the inverse of e modulo $\phi(n)$ exists d:

$$d \cdot e \equiv 1 \pmod{\phi(n)} \text{ ----- (1)}$$

Encryption function: $C \equiv M^e \pmod{n}$

Taking exponent d on both sides:

$$C^d \equiv (M^e)^d \pmod{n}$$

From Eq (1):

$$de = 1 + k\phi(n) \text{ where } k \text{ is an integer}$$

Substitute de:

$$C^d \equiv M^{1+k\phi(n)} \pmod{n}$$

Assume $\gcd(M, p) = \gcd(M, q) = 1$

By Fermat's Little Theorem (FLT):

$$M^{p-1} \equiv 1 \pmod{p} \text{ and } M^{q-1} \equiv 1 \pmod{q}$$

$$C^d \equiv M \cdot (M^{p-1})^{k(p-1)}$$

$$C^d \equiv M \pmod{p} \text{ ----- (2)}$$

Similarly:

$$C^d \equiv M \cdot (M^{q-1})^{k(q-1)}$$

$$C^d \equiv M \pmod{q} \text{ ----- (3)}$$

By Eq (2) and (3), using the Chinese Remainder Theorem:

$$C^d \equiv M \pmod{pq}$$

$$M = C^d \pmod{n}$$

Technical Background

RSA Basics

- RSA uses two keys: Public Key (n, e) → encryption

Private Key (n, d) → decryption

- $n = p \times q$ (product of two primes)
- $\phi(n) = (p-1)(q-1)$
- d is chosen such that $d \times e \equiv 1 \pmod{\phi(n)}$

Encryption Formula

- Ciphertext = C
- Plaintext = M

$$C = M^e \pmod{n}$$

- Use public key (e, n) to encrypt

Decryption Formula

- Ciphertext = C
- Plaintext = M

$$M = C^d \pmod{n}$$

- Use private key (d, n) to decrypt

RSA Encryption Example

Encrypt the message **COMPUTER** using the RSA cryptosystem with key (2773, 17).

Solution (step-by-step):

1. Convert Text to Digits

We map each letter to a number according to the Z(26) table:

C - 02, O - 14, M - 12, P - 15, U - 20, T - 19, E - 04, R - 17

So, the digit string is “02 14 12 15 20 19 04 17”

A	B	C	D	E	F	G	H	I	J	K	L	M
00	01	02	03	04	05	06	07	08	09	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

Figure Z(26) Table

2. Split into Blocks

We split the digit string into 4-digit blocks:

0214 → 214

1215 → 1215

2019 → 2019

0417 → 417

These are the plaintext numeric blocks: [214, 1215, 2019, 417]

3. Encrypt Each Block

Use the formula, $C = M^e \bmod n$

$$c_1 = 214^{17} \bmod 2773, 17 = (10001)_2$$

$$x = 1, \text{ power} = 214 \bmod 2773 = 214$$

Using Fast Modular Exponentiation,

$$c_1 = 214^{17} \bmod 2773 = 364$$

		start		end	
j	a(j)	x	power	x	power
0	1	1	214	214	1428
1	0	214	1428	214	1029
2	0	214	1029	214	2328
3	0	214	2328	214	1142
4	1	214	1142	364	-

$$\text{Similarly, } c_2 = 1215^{17} \bmod 2773 = 417$$

$$c_3 = 2019^{17} \bmod 2773 = 1045$$

$$c_4 = 417^{17} \bmod 2773 = 1620$$

The encrypted blocks are: **364 417 1045 1620**

RSA Decryption Example

1. What is the original message encrypted using the RSA system with $p = 47$, $q = 59$,

$n = pq = 2773$, $e = 17$ if the encrypted (cipher) blocks are:

364 417 1045 1620

Solution (step-by-step):

1) Compute n and $\phi(n)$:

$$n = 47 \times 59 = 2773$$

$$\phi(n) = (47 - 1)(59 - 1) = 46 \times 58 = 2668$$

2) Find d such that $d \times e \equiv 1 \pmod{\phi(n)}$.

We need $17 \times d \equiv 1 \pmod{2668}$.

Using Euclidean Algorithm,

$$2668 = 17 * 156 + 16$$

$$17 = 16 * 1 + 1$$

$$1 = 17 - 1 * 16$$

$$= 17 - 1 * (2668 - 156 * 17)$$

$$= -1 * 2668 + 157 * 17$$

The modular inverse is $d = 157$.

(Because $17 \times 157 = 2669 \equiv 1 \pmod{2668}$).

3) Decrypt each ciphertext block c by computing given ciphertexts: 364 417 1045 1620.

$$364^{157} \bmod 2773, 157 = (10011101)_2$$

$$x = 1, \text{ power} = 364 \bmod 2773 = 364$$

Using Fast Modular Exponentiation,

$$m_1 = 364^{157} \bmod 2773 = 214$$

$$\text{Similarly, } m_2 = 417^{157} \bmod 2773 = 1215$$

$$m_3 = 1045^{157} \bmod 2773 = 2019$$

$$m_4 = 1620^{157} \bmod 2773 = 417$$

		start		end	
j	a(j)	x	power	x	power
0	1	1	364	364	2165
1	0	364	2165	364	855
2	1	364	855	644	1726
3	1	644	1726	2344	874
4	1	2344	874	2182	1301
5	0	2182	1301	2182	1071
6	0	2182	1071	2182	1792
7	1	2182	1792	214	-

A	B	C	D	E	F	G	H	I	J	K	L	M
00	01	02	03	04	05	06	07	08	09	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

Figure Z(26) Table

4) Convert numeric blocks back to letters using Z(26) table:

$$214 \rightarrow 02 \ 14 \rightarrow C \ O$$

$$1215 \rightarrow 12 \ 15 \rightarrow M \ P$$

$$2019 \rightarrow 20 \ 19 \rightarrow U \ T$$

$$417 \rightarrow 04 \ 17 \rightarrow E \ R$$

Concatenate $\rightarrow$ COMPUTER

Final Answer:

The decrypted message is COMPUTER (8 letters)

Suppose that Alice and Bob have these public keys and corresponding private keys:

$$(n_{\text{Alice}}, e_{\text{Alice}}) = (2867, 7) = (61 \cdot 47, 7), d_{\text{Alice}} = 1183$$

and

$$(n_{\text{Bob}}, e_{\text{Bob}}) = (3127, 21) = (59 \cdot 53, 21), d_{\text{Bob}} = 1149.$$

No. 31 from Discrete Mathematics and its Applications

Alice wants to send to all her friends, including Bob, the message "SELL EVERYTHING" so that he knows that she sent it. What should she send to her friends, assuming she signs the message using the RSA cryptosystem.

Given:

$$(n_{\text{Alice}}, e_{\text{Alice}}) = (2867, 7) = (61 \cdot 47, 7)$$

$$d_{\text{Alice}} = 1183$$

$$(n_{\text{Bob}}, e_{\text{Bob}}) = (3127, 21) = (59 \cdot 53, 21)$$

$$d_{\text{Bob}} = 1149$$

Given message: SELL EVERYTHING

A	B	C	D	E	F	G	H	I	J	K	L	M
00	01	02	03	04	05	06	07	08	09	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

Figure Z(26) Table

18 04 11 11 04 21 04 17 24 19 07 08 13 06

Regroup in blocks of 4 digits:

1804 1111 0421 0417 2419 0708 1306

Alice then decrypts each block using the decryption transformation

$$D_{(2867,7)} = D_{(n,e)} = x^d \bmod n = x^{1183} \bmod 2867$$

$$\mathbf{1804^{1183} \bmod 2867 = 2186}$$

$$\mathbf{1111^{1183} \bmod 2867 = 2087}$$

$$\mathbf{0421^{1183} \bmod 2867 = 1279}$$

$$\mathbf{0417^{1183} \bmod 2867 = 1251}$$

$$\mathbf{2419^{1183} \bmod 2867 = 0326}$$

$$\mathbf{0708^{1183} \bmod 2867 = 0816}$$

$$\mathbf{1306^{1183} \bmod 2867 = 1948}$$

The message that Alice sends then consists of the decryption of the blocks:

2186 2087 1279 1251 0326 0816 1948

Bob and friends should decrypt using Alice's public key to confirm the message was sent by Alice and the message has not been corrupted.

No. 32 from Discrete Mathematics and its Applications

Alice wants to send to Bob the message “BUY NOW” so that he knows that she sent it and so that only Bob can read it. What should she send to Bob, assuming she signs the message and then encrypts it using Bob’s public key?

For only Bob to be able to read the message, Alice should encrypt the message using Bob's public key $(n_{\text{Bob}}, e_{\text{Bob}}) = (3127, 21) = (59.53, 21)$

Encoding with $Z(26)$ table,

Message : **“BUY NOW”**

Mapping : **B=01, U = 20, Y = 24, N=13, O = 14 , W = 22**

Digits : **01 20 24 13 14 22**

Block formation,

Blocks (4 digits each): **0120 2413 1422**

Plaintext integers: **$m_1 = 120$, $m_2 = 2413$, $m_3 = 1422$**

Encrypt each block with Bob's key,

$$C_1 = 120^{21} \bmod 3127 = 2711$$

$$C_2 = 2413^{21} \bmod 3127 = 2080$$

$$C_3 = 1422^{21} \bmod 3127 = 280$$

Final ciphertext, 2711 2080 280

Bob should decrypt the message using his private key. This ensures no one else can read the message except Bob.

Application Area

1. Secure File Sharing

RSA is widely used to protect files when they are shared between users or systems. By encrypting files with the recipient's public key, RSA ensures that only the intended person with the corresponding private key can open and read the content. This prevents hackers or unauthorized users from accessing sensitive data during transfer. In organizations, secure file sharing helps protect confidential documents such as contracts, research data, and business plans.

2. Secure Email Communications

Email is one of the most common communication tools, but it is also vulnerable to interception. RSA is used in email encryption systems like PGP (Pretty Good Privacy) to make sure that only the intended recipient can read the message. RSA also provides digital signatures for emails, which confirm the identity of the sender and prevent tampering. This builds trust and security in personal, business, and government communication.

3. VPN and Network Security

Virtual Private Networks (VPNs) rely on RSA to establish secure connections between a user's device and the network. RSA is used during the authentication and key exchange process, ensuring that the session is encrypted and safe from eavesdropping. This is especially important for remote workers or people connecting through public Wi-Fi, as RSA helps keep data traffic private and secure.

4. Confidential Communications in Government and Military

Government agencies and the military handle extremely sensitive information that must remain secure. RSA plays an important role in encrypting communications to prevent espionage, cyberattacks, or leaks. It ensures that classified information, intelligence reports, and strategic plans are accessible only to authorized personnel. RSA's strength makes it suitable for protecting national security.

5. E-commerce Transactions

Online shopping and banking depend heavily on RSA for security. When a customer enters their credit card details on a website, RSA helps encrypt the data before it is sent to the server. This prevents cybercriminals from stealing payment information. RSA also enables secure connections through HTTPS, giving customers confidence that their transactions are safe. Without RSA and similar systems, trust in online commerce would be impossible.

6. Cloud Storage Encryption

Many people and companies store their data in the cloud, which raises concerns about privacy and security. RSA helps secure cloud storage by encrypting files before they are uploaded. Even if hackers gain access to the storage servers, the encrypted files remain unreadable without the proper private key. This ensures that personal data, financial records, and business information remain safe in cloud environments.

7. Internet of Things (IoT) Security

IoT devices, such as smart home systems, medical sensors, and connected cars, are often vulnerable to cyberattacks because of their constant internet connectivity. RSA provides a way to authenticate devices and encrypt the data they send and receive. For example, a smart lock using RSA can verify that commands are coming from the real owner's phone, not from a hacker. By adding RSA-based security, IoT devices become more trustworthy and resistant to attacks.

SecureMail website

Our team has created a frontend-based web application that utilizes the RSA cryptosystem to encrypt and decrypt messages.

[Link to Page](#)

Users can select prime numbers p and q , along with an exponent e . If e is not relatively prime to $\phi(n) = (p-1)(q-1)$, an alert box will notify the user.

The screenshot shows a web application titled "EMAIL ENCRYPTION MECHANISM". It features a "SECURE MESSAGE COMPOSER" section with input fields for "YOUR NAME" (Group1), "RECIPIENT NAME" (Tr. Myat), and "SUBJECT" (Secure Message). To the right, the "CRYPTOGRAPHIC PARAMETERS" section allows selection of "PRIME P" (43), "PRIME Q" (59), and "EXPONENT E" (13). A note states "Must be coprime with $\phi(n)$ ". Below this, the "GENERATED KEYS" section displays "PUBLIC KEY (N)", "EXPONENT (E)", and "PRIVATE KEY(D)". At the bottom, a "MESSAGE" section has a "MESSAGE TO ENCRYPT" field and a "PROCESS MESSAGE" button. "ENCRYPT" and "DECRYPT" buttons are also present at the top right of the composer.

Once the three values are chosen and a message is entered in the "Message to Encrypt" section, clicking the "Process Message" button will automatically generate the private key d , as well as display the numeric and text outputs.

This close-up shows the "CRYPTOGRAPHIC PARAMETERS" section with "PRIME P" set to 53, "PRIME Q" set to 59, and "EXPONENT E" set to 3. The "GENERATED KEYS" section below it shows the calculated values: "PUBLIC KEY (N)" is 3127, "EXPONENT (E)" is 3, and "PRIVATE KEY(D)" is 2011. A note "Must be coprime with $\phi(n)$ " is visible between the parameter and key sections.

MESSAGE

MESSAGE TO ENCRYPT

STOP

NUMERIC OUTPUT

2422 1946

TEXT OUTPUT

YWT0

PROCESS MESSAGE

To decrypt, users can copy the numeric output, switch to decrypt mode, paste the values into the "Encrypted Numbers" input box, and process the message to reveal the original text.

MESSAGE

ENCRYPTED NUMBERS

2422 1946

NUMERIC OUTPUT

1819 1415

TEXT OUTPUT

STOP

PROCESS MESSAGE

The mini-project includes the following functions:

1. The Extended Euclidean Algorithm, which is used to compute the private key (d).
2. A modular exponentiation function that takes three parameters: base, exponent, and modulus ($B^e \bmod n$). This function is used for both encryption (M, e, n) and decryption (C, d, n) through fast modular exponentiation.

Conclusion

RSA is one of the most widely known and trusted public-key cryptosystems. Its design is based on strong mathematical principles, including modular exponentiation, Fermat's Little Theorem (FLT), and the Extended Euclidean Algorithm, which make encryption and decryption both practical and secure. These mathematical foundations ensure that RSA remains difficult to break, even with powerful computers, as long as sufficiently large key sizes are used.

Over the years, RSA has become an essential part of modern digital security. Its applications span many fields such as secure banking transactions, e-commerce payments, email protection, digital identity verification, and the safeguarding of sensitive communications. Beyond everyday consumer uses, RSA also plays a critical role in government, military, and cloud security, where the confidentiality and authenticity of information are of utmost importance.

In summary, RSA has not only solved the long-standing problem of secure key distribution but has also laid the foundation for trust in the digital world. Even though newer cryptographic systems are being developed, RSA continues to serve as a cornerstone of cybersecurity, providing confidentiality, authentication, and integrity across countless applications that shape how people safely interact online today.

Reference

- Discrete Mathematics and Its Application, Kenneth H Rosen, 8th Edition, ISBN (13 digits):
978-0-07-068188-0